3 Transformations for 3D movement

We can break down $\vec{P_{3D}}$ into 2 states: $\vec{P_{3DI}}$ and $\vec{P_{3DF}}$. $\vec{P_{3DI}}$ represents the location of a point before a set of transformations, $\{T_0, T_1, \ldots, T_n\}$, that describe its movement. $\vec{P_{3DF}}$ represents the location of a point after the set of transformations.

Let $T_n : \mathbb{R}^3 \to \mathbb{R}^3$.
Let T_n be of the form $\left\{T_n(step) \middle| \vec{P_{3DF}} = \mathbf{A}\vec{P_{3DI}} + \vec{b}\right\}$

Keypresses are polled for 1 or more of the following transformations $\{T_{rx}(\Delta\theta), T_{ry}(\Delta\theta), T_{rz}(\Delta\theta), T_z(\Delta z)\}$, which apply to all points during each iteration of the render loop. The table below describes the function and expression of each transformation.

Our POV	Point Movement	T_n	Expression
Move forward	Lower Z-component value	$T_z(-\Delta z)$	$\vec{P_{3DF}} = \mathbf{I_3}\vec{P_{3DI}} - \Delta z\hat{z}$
Move backward	Raise Z-component value	$T_z(+\Delta z)$	$\vec{P_{3DF}} = \mathbf{I_3}\vec{P_{3DI}} + \Delta z\hat{z}$
Pitch up	Rotate around x-axis counter-clockwise	$T_{rx}(+\Delta\theta)$	$\vec{P_{3DF}} = \mathbf{R_x}(+\Delta\theta)\vec{P_{3DI}} + \vec{P_{proj}}$
Pitch down	Rotate around x-axis clockwise	$T_{rx}(-\Delta\theta)$	$\vec{P_{3DF}} = \mathbf{R_x}(-\Delta\theta)\vec{P_{3DI}} + \vec{P_{proj}}$
Yaw left	Rotate around y-axis counter-clockwise	$T_{ry}(+\Delta\theta)$	$\vec{P_{3DF}} = \mathbf{R_y}(+\Delta\theta)\vec{P_{3DI}} + \vec{P_{proj}}$
Yaw right	Rotate around y-axis clockwise	$T_{ry}(-\Delta\theta)$	$\vec{P_{3DF}} = \mathbf{R_y}(-\Delta\theta)\vec{P_{3DI}} + \vec{P_{proj}}$
Roll left	Rotate around z-axis clockwise	$T_{rz}(-\Delta\theta)$	$\vec{P_{3DF}} = \mathbf{R_z}(-\Delta\theta)\vec{P_{3DI}} + \vec{P_{proj}}$
Roll right	Rotate around z-axis counter-clockwise	$T_{rz}(+\Delta\theta)$	$\vec{P_{3DF}} = \mathbf{R_z}(+\Delta\theta)\vec{P_{3DI}} + \vec{P_{proj}}$

Expansions:

$$\vec{P_{proj}} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\hat{z} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\mathbf{I_3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R_x}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & \sin(\theta) \\ 0 & -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$\mathbf{R_y}(\theta) = \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix}$$

$$\mathbf{R_z}(\theta) = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$